PROBLEM CL2

Bore over a Stagnant Water Layer

Original statement · CL2.pdf

Moving frame | Mass and momentum | Hydrostatic pressure | Froude number

OFFICIAL PROBLEM STATEMENT

The bore sketched in the figure of CL2.pdf moves under the action of gravity with constant velocity U over an initially stagnant water layer of thickness h_0, leaving behind a layer of thickness h_1 moving with velocity U_1. Assuming that in the air both viscous stresses and pressure variations with respect to the ambient value p_a are negligible, and so are viscous stresses on the bottom wall, find the values of U_1 and h_1 as a function of U, h_0 and g. As a preliminary computation, obtain the pressure distributions both upstream and downstream from the bore, verifying that they can be expressed in the form $p = p_a + \rho g(h_0 - z)$ and $p = p_a + \rho g(h_1 - z)$, respectively, where z represents the vertical distance to the bottom wall.

Source: CL2.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eqs. 3.10-3.11, integral momentum balance Eqs. 3.35-3.36.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Planar flow (quantities per unit width normal to the figure) of an incompressible liquid of density ρ, under gravity $\mathbf{g} = -g\,\mathbf{e}_z$. The bore advances towards $-\mathbf{e}_x$ (the x axis points to the right in the figure) with constant velocity U over the stagnant layer of thickness h_0; behind it the layer has thickness h_1 and uniform velocity $-U_1\mathbf{e}_x$.
- Far from the front, in each region, the velocity is horizontal and uniform and the free surface is horizontal at pressure p_a. Viscous stresses are negligible (given); the ratio of viscous to inertial forces is of order $\mu/(\rho U h) \ll 1$ for a water layer.
- The two unknowns are h_1/h_0 and U_1/U; the data are U, h_0, g and ρ.

Symbol	Meaning	SI unit
U	velocity of the bore (relative to the stagnant layer)	m/s
U_1	velocity of the water behind the bore	m/s
h_0, h_1	layer thickness ahead of and behind the bore	m
p_a	ambient pressure acting on the free surface	Pa
g	gravitational acceleration	m/s^2
$F = U^2/(g h_0)$	Froude number of the bore (dimensionless)	1

Dimensional check: $[U^2/(g h_0)] = (\mathrm{m}^2/\mathrm{s}^2)/\big((\mathrm{m}/\mathrm{s}^2)\,\mathrm{m}\big) = 1$.

PHASE 2

Coordinates, Frames and Changes of Variable

- Laboratory frame L: Cartesian basis $(\mathbf{e}_x, \mathbf{e}_z)$ with z measured from the bottom wall. The flow is unsteady in L because the front moves.
- Frame of the bore B: it moves with the velocity $-U\,\mathbf{e}_x$ relative to L. Since this velocity is constant, B is inertial and no inertial forces appear in the balances. Velocities in B are $\mathbf{v}_B = \mathbf{v}_L + U\,\mathbf{e}_x$: the liquid ahead has $\mathbf{v}_B = U\,\mathbf{e}_x$ and the liquid behind has $\mathbf{v}_B = (U - U_1)\,\mathbf{e}_x$. In B the front is stationary, so the flow is steady.
- Control volume fixed in B (so $\mathbf{v}_c = 0$; Eqs. 3.11 and 3.36): Σ_0 (vertical section ahead of the front, $\mathbf{n} = -\mathbf{e}_x$, area h_0 per unit width), Σ_1 (vertical section behind, $\mathbf{n} = +\mathbf{e}_x$, area h_1), Σ_b (bottom wall, $\mathbf{n} = -\mathbf{e}_z$) and Σ_s (free surface, material and steady in B). On Σ_b and Σ_s, $\mathbf{v}\cdot\mathbf{n} = 0$.

PHASE 3

Step-by-Step Derivation

3.1 Preliminary result: hydrostatic pressure upstream and downstream

Why this tool: in the uniform regions the velocity is horizontal, so the z component of the integral momentum balance (Notes Eq. 3.36) has no momentum flux and reduces to a force balance. Take a fixed control volume formed by the liquid column of width Δx (per unit width) between a height z and the free surface $z = h_0$ (upstream region). Its faces are: the bottom face at height z ($\mathbf{n} = -\mathbf{e}_z$), the free surface ($\mathbf{n} = +\mathbf{e}_z$, $p = p_a$) and two vertical faces ($\mathbf{n} = \mp\mathbf{e}_x$).

The z component of $\rho\,\mathbf{v}\,(\mathbf{v}\cdot\mathbf{n})$ vanishes on every face because $v_z = 0$. The pressure force has z component $+p(z)\Delta x$ on the bottom face and $-p_a\Delta x$ on the free surface; the pressure on the vertical faces points along $\pm\mathbf{e}_x$ and viscous stresses are negligible. The weight of the column is $-\rho g\,(h_0 - z)\,\Delta x$. Hence

$$0 = p(z)\,\Delta x - p_a\,\Delta x - \rho g\,(h_0 - z)\,\Delta x \implies \boxed{p = p_a + \rho g\,(h_0 - z)} \quad \text{(upstream)}.$$

The same argument with the surface at $z = h_1$ gives $p = p_a + \rho g\,(h_1 - z)$ downstream.

3.2 Mass balance in the frame of the bore

Why this tool: steady flow and fixed control volume in B, so $\int_{\Sigma_c}\rho\,\mathbf{v}\cdot\mathbf{n}\,d\sigma = 0$ (Notes Eq. 3.11). Only Σ_0 and Σ_1 are crossed by the flow:

$$-\rho\,U\,h_0 + \rho\,(U - U_1)\,h_1 = 0 \implies U\,h_0 = (U - U_1)\,h_1. \qquad \text{(c1)}$$

3.3 Momentum balance in the frame of the bore

Why this tool: the unknown h_1 appears through the pressure forces, which only the momentum balance (Notes Eq. 3.36) contains. Since $\int_{\Sigma_c} p_a\,\mathbf{n}\,d\sigma = 0$, replace p by $p - p_a$. The x component of the left-hand side is

$$\int_{\Sigma_c}\rho\,v_x\,(\mathbf{v}\cdot\mathbf{n})\,d\sigma = \rho U\,(-U h_0) + \rho\,(U - U_1)\,(U - U_1)\,h_1 = -\rho U^2 h_0 + \rho\,(U - U_1)^2\,h_1.$$

Right-hand side: on Σ_s, $p - p_a = 0$; on Σ_b the normal is $-\mathbf{e}_z$, with no x component; the viscous stresses are negligible (hypothesis); gravity has no x component. The pressure on Σ_0 and Σ_1 is the hydrostatic distribution of 3.1, with $\mathbf{n} = -\mathbf{e}_x$ and $+\mathbf{e}_x$ respectively. The antiderivative of $\rho g\,(h - z)$ is $\rho g\,(hz - z^2/2)$ and Barrow's rule between $z = 0$ and $z = h$ gives $\rho g\,(h^2 - h^2/2) = \rho g h^2/2$:

$$-\int_{\Sigma_0}(p - p_a)\,\mathbf{n}\,d\sigma\cdot\mathbf{e}_x = +\int_0^{h_0}\rho g\,(h_0 - z)\,dz = \frac{\rho g h_0^2}{2}, \qquad -\int_{\Sigma_1}(p - p_a)\,\mathbf{n}\,d\sigma\cdot\mathbf{e}_x = -\int_0^{h_1}\rho g\,(h_1 - z)\,dz = -\frac{\rho g h_1^2}{2}.$$

$$-\rho U^2 h_0 + \rho\,(U-U_1)^2\,h_1 = \frac{\rho g}{2}\left(h_0^2 - h_1^2\right). \qquad \text{(c2)}$$

3.4 Solution for h_1/h_0 and U_1/U

From (c1), $U - U_1 = U h_0/h_1$, so $(U-U_1)^2\,h_1 = U^2 h_0^2/h_1$. Substituting into (c2) and dividing by ρ:

$$U^2 h_0\left(\frac{h_0}{h_1} - 1\right) = \frac{g}{2}\,(h_0 - h_1)(h_0 + h_1) \implies U^2 h_0\,\frac{h_0 - h_1}{h_1} = \frac{g}{2}\,(h_0 - h_1)(h_0 + h_1).$$

For a genuine bore $h_1 \neq h_0$ (the case $h_1 = h_0$ is the undisturbed layer), so the factor $h_0 - h_1$ can be cancelled. With $r \equiv h_1/h_0$ and $F = U^2/(g h_0)$, dividing by $g h_0^2$ gives $F/r = (1+r)/2$, that is

$$r^2 + r - 2F = 0 \implies \boxed{\frac{h_1}{h_0} = \frac{1}{2}\left(\sqrt{1+8F} - 1\right)} \qquad \text{(the negative root is rejected since } h_1 > 0\text{)}.$$

The velocity behind follows from (c1): $U_1/U = 1 - h_0/h_1 = 1 - 1/r = (r-1)/r$. With $s \equiv \sqrt{1+8F}$, $r = (s-1)/2$ and

$$\boxed{\frac{U_1}{U} = 1 - \frac{2}{s-1} = \frac{\sqrt{1+8F} - 3}{\sqrt{1+8F} - 1}}, \qquad F = \frac{U^2}{g h_0}.$$

Erratum in the official solution (reading of the handwriting)

CL2.pdf gives $h_1/h_0 = \frac{1}{2}(\sqrt{1+8F}-1)$, which agrees with the result above. For the velocity it gives, as legible in the scan, $U_1/U = \frac{3}{2}\left(\frac{1}{3}\sqrt{1+8F}-1\right) = (\sqrt{1+8F}-3)/2 = r-1$. This contradicts the official continuity equation $Uh_0 = (U-U_1)h_1$, which requires $U_1/U = (r-1)/r$; the two expressions coincide only for $r=1$. The check that proves it: $(U-U_1)h_1 = Uh_0 > 0$ forces $U_1 < U$, whereas $r-1>1$ for $F>3$ (for $F=4$, $r-1 = 1.3723$ against the correct $U_1/U = 0.5785$, and the mass residual of (c1) with the scan formula is $Uh_0-(U-U_1)h_1 \neq 0$). The computed value is kept.

PHASE 4

Interpretation, Limits and Dimensional Check

- The bore exists only for $F>1$: $r>1 \iff s>3 \iff F>1$. F is the Froude number of the front relative to the shallow-water wave speed $\sqrt{gh_0}$; a front slower than $\sqrt{gh_0}$ cannot raise the level.
- Weak bore, $F = 1+\epsilon$: $r \approx 1+\frac{2}{3}\epsilon$ and $U_1/U \approx \frac{2}{3}\epsilon \to 0$; the front degenerates into an infinitesimal gravity wave travelling at $\sqrt{gh_0}$.
- Strong bore, $F \gg 1$: $r \approx \sqrt{2F}$, so $h_1 \approx h_0\sqrt{2}\,U/\sqrt{gh_0}$ and $U_1 \to U$: the liquid behind moves almost with the bore.
- Dimensions (SI): both sides of (c2) are forces per unit width, $[\rho U^2 h_0] = [\rho g h^2] = \mathrm{N/m}$; F, r and U_1/U are dimensionless.
- Numerical cross-check: $F=4$, $h_0 = 0.5$ m, $g = 9.81$ m/s^2 give $U = 4.4294$ m/s, $h_1/h_0 = 2.3723$ ($h_1 = 1.1861$ m), $U_1/U = 0.57846$ ($U_1 = 2.5623$ m/s); the residuals of (c1) and (c2) are below 10^{-14} in SI units.

Extension (beyond the official solution)

Mechanical energy is not conserved across the bore: Bernoulli's equation between both sections would give a different relation. The head lost is the classical hydraulic-jump value $\Delta H = (h_1-h_0)^3/(4h_0h_1)$, always positive for $h_1 > h_0$.